MIL-B-13889C(WC)
9 May 1973
SUPERSEDING
MIL-B-13889B(WC)
15 April 1970

MILITARY SPECIFICATION

BRUSHES AND STAFFS, CANNON BORE

1. SCOPE

1.1 This specification covers brushes, wood staffs, and metal staffs for coupling to the brushes for cleaning cannon bores (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

SPECIFICATIONS

Military
MIL-C-13931 - Cannons, General Specification for
MIL-P-14232 - Parts, Equipment and Tools for Army Materiel, Packaging and Packing of

STANDARDS

Federal
FED-STD-595 - Colors

Military
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-130 - Identification Marking of U.S. Military Property

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 1090

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3. REQUIREMENTS

3.1 First article. When specified in the contract, the contractor shall submit a first article. Unless otherwise specified, (see 6.2), the first article shall be included in the pilot pack (see 5.1).

3.2 Design. The brushes and staffs shall conform to the design specified herein, to the applicable drawings (see 6.2) and to MIL-C-13931.

3.2.1 Alternative design. When alternative designs are shown on the applicable drawings, the supplier shall notify the contracting officer, prior to production, which design will be used.

3.3 Materials and construction. The brushes and staffs shall conform to the material and construction requirements specified herein, the applicable drawings (see 6.2) and to MIL-C-13931.

3.3.1 Alternative materials. When alternative materials are indicated on the applicable drawings, the supplier shall notify the contracting officer, prior to production, which materials will be used.

3.3.2 Brushes.

3.3.2.1 Bristle distribution. The loading of bristles per unit of length of brush strip shall be uniform. Where more than one type of bristle is specified, the metallic and non-metallic bristles shall be uniformly mixed and distributed on the strips.

3.3.2.2 Bristle retention. The bristles shall be mechanically secured in the strip so that a bristle will withstand a longitudinal pull of not less than eight ounces.

3.3.3 Staff.

3.3.3.1 Coupling attachment. The coupling and staff joints shall withstand the load specified on the applicable drawing without perceptible loosening or failure.

3.3.3.2 Straightness. The deviation from straightness of a staff section shall not exceed 0.04 inch per foot for the length of the section.

3.3.3.3 Color. When painting is required by the drawings, staffs shall be of a uniform shade of Color Chip Number 34052, 34079, 34086, 34087, 34096 or 34102 of Federal Standard 595.

3.4 Marking. Marking of brushes and staffs shall be in accordance with the applicable drawings and marking provisions of MIL-STD-130.

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3.5 Workmanship. Workmanship shall be in accordance with MIL-C-13931.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions used herein are in accordance with MIL-STD-109.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)

b. Quality conformance inspection (see 4.5)

4.4 First article inspection. A first article shall be submitted for inspection in accordance with contract requirements. The first article shall be representative of the production processes to be used during quantity production. The first article shall be subjected to the quality conformance inspections specified herein and such other inspection as is necessary to determine compliance with the requirements of the contract.

4.5 Quality conformance inspection.

4.5.1 Inspection performance. Inspection shall be performed in accordance with MIL-C-13931 except as otherwise specified herein.

4.5.2 Sampling inspection. When sampling inspection is specified herein, procedures shall be in accordance with MIL-C-13931.

4.5.3 Examinations.

4.5.3.1 Bristle distribution. Each brush shall be examined to determine that the specified quantity of metallic and non-metallic bristles are uniformly mixed and distributed on the strips.

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4.5.3.2 Staff section straightness. Each assembled staff section shall be examined for straightness with a flat surface plate and feeler gage to determine compliance with 3.3.3.2. The staff section shall be positioned horizontally on the surface plate and rolled so that it rotates 360°. The deviation from straightness shall be determined by measuring the largest gap opening between the top of the surface plate and the bottom of the staff with the feeler gage.

4.5.3.3 Welds. Where welding is specified, each staff section shall be inspected for completeness, appearance and conformance with the requirements specified on the applicable drawings.

4.5.3.4 Surface texture. The surface texture shall be examined as specified in MIL-C-13931 to determine conformance with the requirements of the applicable drawings.

4.5.3.5 Color. Painted staffs shall be visually examined for uniformity of color in accordance with 3.3.3.3.

4.5.3.6 Classification of characteristics and acceptable quality levels (AQL). Where specified herein, classification of characteristics and AQLs shall be in accordance with MIL-C-13931.

4.5.3.6.1 End caps.

Class	Characteristic	AQL	Inspection Method
Minor			
201	Cracks, blisters, warped or chipped surfaces	2.5	Visual
202	Dimensional requirements	4.0	SME
203	Workmanship	4.0	Visual

4.5.3.6.2 Adapters.

Minor			
201	Thread requirements	2.5	SME/Visual
202	Dimensional requirements	4.0	SME
203	Workmanship	4.0	Visual

4.5.3.6.3 Center pins and eyebolts.

Minor			
201	Diameter and centrality of pin hole	2.5	SME
202	Dimensional requirements	2.5	SME
203	Workmanship	4.0	Visual

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4.5.3.6.4 Brushes.

Class	Characteristics	AQL	Inspection method
Minor			
201	Dimensional requirements	2.5	SME
202	Brush trim requirements	4.0	Visual
203	Workmanship	4.0	Visual

4.5.3.6.5 Bore brush assembly.

Minor			
201	Marking requirement	2.5	Visual
202	Completeness and security of assembly	2.5	Manual/Visual
203	Workmanship	4.0	Visual

4.5.3.6.6 Staff sections.

Minor			
201	Marking requirements	2.5	Visual
202	Thread requirements	2.5	SME/Visual
203	Dimensional requirements	4.0	SME
204	Workmanship	4.0	Visual
205	Color	4.0	Visual

4.5.3.7 Final examination. Final examination of each bore brush and staff shall be completed after testing and prior to preservation and packaging to determine compliance with this specification, with MIL-C-13931 and with the contract.

4.5.3.8 Inspection of preparation for delivery.

4.5.3.8.1 Packaging inspection. Unless otherwise specified, inspection to determine compliance with cleaning, preservation, packaging, packing and marking requirements of the applicable packaging data sheets for the level designated in the contract shall be as specified in MIL-P-14232.

4.5.4 Tests.

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4.5.4.1 Bristle retention. A sample, randomly selected from a lot at an AQL of 2.5, shall be tested as follows: Each coil of brush strip shall be tested by applying a tensile force of 8 ounces to an individual bristle, in a direction along the axis of the bristle and normal to the axis of the coil. The device employed to grip the bristle shall be designed to hold the required load by vise action, with a minimum of bristle distortion. The required load shall be applied through, or resisted by, a scale or weight balance capable of indicating when the required load on the bristle is obtained. The load shall be applied gradually at a rate not exceeding one ounce per second. Test of a brush coil shall be performed on a bristle near each end of the strip, and on one near the middle of the strip length. If the bristle being pulled breaks, retest shall be made on another bristle in approximately the same location. Extraction of any of the tested bristles by the specified pull shall be cause for rejection of the brush coil.

4.5.4.2 Staff coupling attachment. A sample of assembled staff sections selected randomly from a lot at an AQL of 2.5 shall be tested as follows: Each staff section shall be tested for compliance with the requirements of the drawing for the strength of the coupling attachment. Suitable testing equipment of a mechanical, hydraulic or other type with an appropriate mechanical advantage, shall apply the specified load for not less than 30 seconds. After the test, each staff section shall be examined visually and manually. Any looseness or visible damage to the coupling joints shall be cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Pilot pack. On each contract, a pilot pack (see 6.5) shall be forwarded in accordance with 3.1. Pilot packs for the cannon shall be packaged to the level of packaging specified in the contract and packed level C in accordance with the requirements of the applicable packaging data sheet (see 6.1).

5.2 Preservation, packaging, and marking. The brushes and staffs shall be unit packaged, packed, and marked in accordance with the requirements of the applicable packaging data sheet for the level of protection specified in the contract (see 6.2.1).

5.3 Repair parts. Repair parts shall be unit packaged, packed, and marked in accordance with the requirements of the applicable packaging data sheet (see 6.2.1).

6. NOTES

6.1 Intended use. This specification is intended for, but not limited to, procurement of the following items:

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Intended use cont'd.

FSN	DWG. NO.	DESCRIPTION	PACKAGING DATA SHEET
1005-650-8210	D8766012	Staff, Cleaning (20mm)	P8766012
1010-616-3574	C6163574	Staff, Cleaning M9	P6163574
1010-619-3748	B6193748	Brush Section Cleaning Artillery	P6193748
1010-619-3770	B6193770	Brush, Bore M18 (57mm)	P6193770
1010-624-0944	B6240944	Brush, Bore M29 (40mm)	P6240944
1010-819-4516	B8768250	Brush, Bore Assembly (37mm)	P8768250
1015-557-0617	C5570617	Staff, Cleaning M8	P5570617
1015-615-7208	C6157208	Brush, Section Cleaning Artillery	P6157208
1015-615-7220	C6157220	Brush, Section Cleaning Artillery	---
1015-616-8009	C6168009	Brush, Bore M12 (105mm)	P6168009
1015-616-8031	C6168031	Brush, Bore M15 (76mm)	P6168031
1015-616-8038	C6168038	Brush, Bore M9 (75mm)	P6168038
1015-618-1971	C6181971	Brush, Section Cleaning Artillery	P6181971
1015-618-1980	C6181980	Brush, Bore M19 (90mm)	P6181980
1015-618-1983	C6181983	Brush, Section Cleaning Artillery	---
1015-619-7201	C6197201	Brush, Section Cleaning Artillery	---
1015-619-7207	C6197207	Brush, Bore M21 (120mm)	P6197207
1015-678-7110	C8766039	Brush, Bore Assy (105mm & 106mm)	P8766039
1015-679-5593	C8766032	Brush, Bore Assy (90mm)	P8766032
1015-783-8790	C8765990	Brush, Section, Cleaning Artillery	---
1015-793-1994	C8765992	Brush, Bore Assembly (120mm)	P8765992
1015-857-0421	C8768698	Brush, Bore Assembly (4.2 In.)	P8768698

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FSN	DWG. NO.	DESCRIPTION	PACKAGING DATA SHEET
1015-859-4516	C8767959	Brush, Bore Assembly (90mm)	P8767959
1025-112-8552	C11578061	Brush, Bore Assembly (152mm)	P11578061
1025-113-2968	C11578060	Brush, Section Cleaning Artillery	P11578060
1025-239-9639	C8769220	Brush, Bore Assembly (165mm)	P8769220
1025-591-6734	D8769219	Brush, Section Cleaning Artillery	P8769219
1025-730-7400	C8765994	Brush, Bore Assembly (155mm)	P8765994
1025-783-8791	C8765991	Brush, Section Cleaning Artillery	P8765991
1025-793-1990	C7227121	Brush, Section Cleaning Artillery	P7227121
1025-857-0322	C8767430	Brush, Bore Assembly (155mm)	P8767430
1025-862-9092	C8767935	Brush, Bore Assembly (175mm)	P8767935
1025-862-9094	D8767938	Brush, Section Cleaning Artillery	P8767938
1025-918-5476	D8769852	Brush, Bore and Groove (152mm)	P8769852
1025-981-1258	B8768252	Brush, Section Cleaning Artillery	---
1030-532-3857	C5573731	Brush, Section Cleaning Artillery	P5573731
1030-537-3715	C8766350	Brush, Bore Assembly (8 In.)	P8766350
1090-699-0633	D7309259	Staff, Section	P7309259
1090-563-7232	D7309228	Staff, Section M15	P7309228
----	B5279075	Brush, Section Cleaning Artillery	P5279075
----	B6110689	Staff, Cleaning Assembly	P6110689
----	B6194450	Staff, Middle	P6194450
----	B8767957	Brush Section Cleaning Artillery	----

6.2 Ordering data. Procurement documents should specify the following:

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6.2.1 Procurement requirements.

a. Title, number and date of this specification

b. Lists of drawings and specifications pertinent to the brushes and staffs, showing applicable revision dates.

c. The applicable packaging data sheets (see 6.1) and the levels of packaging and packing.

d. The availability of inspection equipment from Government stock.

e. The extent of supplier's responsibility for Government-furnished and supplier-acquired final inspection equipment.

f. The requirements for submission of, and shipping instructions for, a first article (see 3.1 and 4.4) and pilot pack if different (see 3.1 and 5.1).

6.2.2 Contract data requirements. Monthly reports of the results of final examination and performance testing shall be as specified for delivery on DD Form 1423 in the contract.

6.3 When warranted, the contract should specify the application of MIL-Q-9858 or MIL-I-45208, as appropriate, on the Management Control Systems Summary List, DD Form 1660.

6.4 Unless otherwise specified, the contract should specify the application of MIL-I-45607 and MIL-C-45662 on the Management Control Systems Summary List, DD Form 1660.

6.5 Definition of pilot pack. A requirement in conjunction with a specification or contract for a first article in order to base approval of packages.

Custodian:
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Preparing activity:
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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